## Example 7.4

In how many ways, can the students answer 3 true or false type questions in a slip test?

***Solution:***

(i) Assuming that the question $Q_1$ is answered True, questions $Q_2$ and $Q_3$ can be answered as **TT, TF, FT** and **FF** in 4 ways.

(ii) Assuming that the question $Q_1$ is answered False, $Q_2$ and $Q_3$ can also be answered as **TT, TF, FT,** and **FF** in 4 ways.

Thus, as each question has only two options (True or False) in 2 ways, the number of ways of answering these 3 questions in a slip test is **8 (2 x 2 x 2) possible ways** as shown in Fig. 7.8.

**Number of ways of answering the questions**

| | Questions | | |
|---|---|---|---|
| | $Q_1$ | $Q_2$ | $Q_3$ |
| | T/F | T/F | T/F |
| Answer1 | T | T | T |
| Answer2 | T | T | F |
| Answer3 | T | F | T |
| Answer4 | T | F | F |
| Answer5 | F | T | T |
| Answer6 | F | T | F |
| Answer7 | F | F | T |
| Answer8 | F | F | F |

$[(1\ (Q_1) \times 2\ (T/F)) \times (1\ (Q_2) \times 2\ (T/F)) \times (1\ (Q_3) \times 2\ (T/F))] = 8$ ways

Fig. 7.8

## Example 7.5

Madhan wants to a buy a new car. The following choices are available for him.

- There are **2 types** of cars as shown in the Fig. 7.9
- There are **5 colours** available in each type as shown in Fig. 7.9.
- There are **3 models** available in each colour
  - (i) **GL** (standard model)
  - (ii) **SS** (sports model)
  - (iii) **SL** (luxury model)

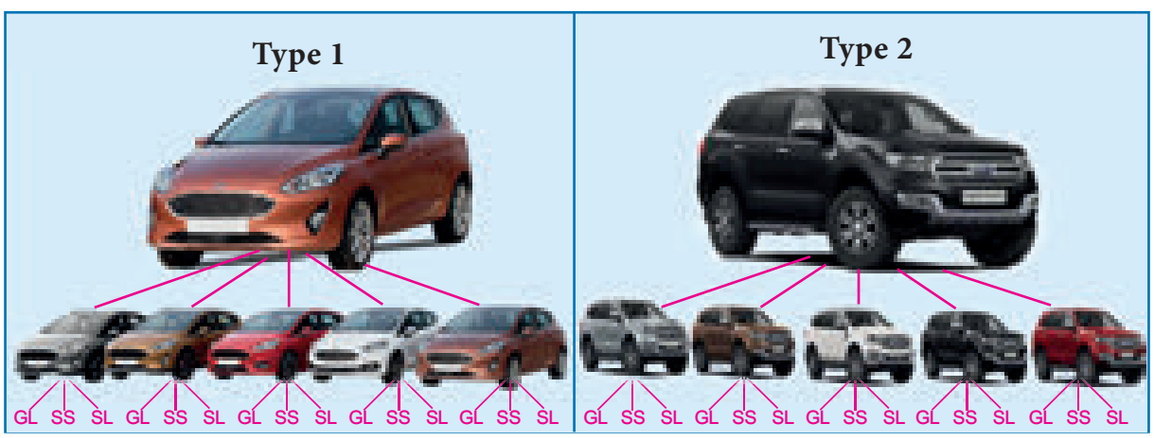



Fig. 7.9

(i) In how many different ways can Madhan buy any one of the new car?

(ii) If the **white colour** is not available in **Type 2**, then in how many ways can Madan buy a new car among the given option?



07_8th_Maths_IP_Chap 7.indd 238 2/12/2020 12:21:48 PM